

■ CortexKPI — Autonomous Executive Storytelling Engine

Accenture Innovation Challenge 2026 | Problem Statement 3: BusinessIntelligence.ai

Team: Team cognition (IIT Jodhpur) | Authors: Abhishek Gehlot & Saurav Gupta

1. Executive Overview & Problem Statement

When high-stakes enterprise metrics drop (e.g. an unexpected **\$250,000/day revenue loss** in APAC), executive leadership and engineering teams spend 6+ hours sifting through fragmented Datadog APM charts, Jira pull requests, Zendesk support tickets, and Slack alerts trying to answer: *What happened, why it happened, and how to fix it?*

CortexKPI is a fully production-dynamic AI diagnostic engine that unifies seasonality-adjusted Bayesian anomaly detection, causal DAG decomposition with generalized graph propagation, multimodal RAG log vectorization, and epistemic safeguards into an interactive executive dashboard with **1-Click automated mitigation**. Zero hardcoded values — the entire 4-layer ML pipeline adapts to any uploaded dataset automatically.

2. Architectural Pillars (The 4 ML Layers)

Layer / Module	Underlying ML Algorithm	Core Purpose & Enterprise Value
Step 1: Metric Timeline & Anomaly Detection	Seasonal Day-of-Week Adjustment + Bayesian Rolling Baselines (Student-t) + Isolation Forest Unsupervised ML	Seasonality factor adjusts baselines by day-of-week patterns; calculates dynamic
Step 2: Causal Metric Dependency Chain	Structural Causal DAG (NetworkX) + Generalized Edge Propagation + Counterfactual What-If Matrix	Auto-adapts graph to any uploaded KPI schema. Data-driven variance attribution
Step 3: Multimodal RAG Evidence Fusion	TF-IDF N-Gram Vectorizer + Cosine Similarity + Exponential Temporal Decay	Dynamically generates search queries from anomaly context. Weights logs by ten
Step 4: Executive Verdict & 1-Click Fix	4-Pillar Epistemic Safeguards + Dual-Engine Synthesis + Local Qwen 3.8 / LLM AI Hook	Surge-aware narrative (green for growth, red for outage). Context-aware action bu

3. Production-Level Dynamic Features

- **Zero Hardcoded Values:** All fallback defaults derived from live data baselines — no magic numbers (removed 420000, 98.0, 2.85, 120000, 185.0).
- **Seasonality-Adjusted Baselines:** Day-of-week seasonal factors actively adjust rolling means to eliminate false weekend alarms.
- **Auto-Detect Scenario Type:** Dynamically classifies scenarios as GROWTH_SURGE, ANOMALY_DROP, or STABLE from data patterns.
- **Generalized DAG Propagation:** What-If simulator propagates through any parent-child edges, not just hardcoded e-commerce pairs.
- **Memory-Safe File Upload:** Streamed to disk via shutil.copyfileobj — handles enterprise-scale datasets without OOM.
- **Surge-Aware UI:** Green banners, scale-infrastructure buttons, and SURGE technical logs for positive growth scenarios.
- **Robust Error Handling:** All 7 API fetch calls validate HTTP status before JSON parsing.

4. Dual-Mode UX & Persona-Aware Dashboard

- **Simple View vs Technical Diagnostics:** Toggle between plain-English executive summaries and deep statistical Z-score matrices.

- **C-Suite Persona:** Financial impact (~\$250K/day), strategic ROI, and 1-click executive actions.
- **DevOps Persona:** Terminal-style logs, Z-scores, deployment triage, and hotfix rollback buttons.
- **BI Analyst Persona:** Variance attribution %, p-values, cosine similarity scores, and hypothesis rejection logs.
- **Interactive Node Inspector:** Click any metric node to filter evidence logs (with toggle-clear support).
- **What-If Slider:** Real-time percentage delta display (+XX.X%) as you drag any metric.

5. Real-World Datasets Ingested (17,000+ Total Records)

- **metrics_timeseries.csv:** 9,125 rows across 5 global regions (APAC, EU, NA, LATAM, EMEA) over 365 days.
- **jira_deployments.csv:** 20 live-crawled deployment logs from GitHub (Stripe, React, FastAPI, Next.js).
- **zendesk_tickets.csv:** Customer support tickets with NLP sentiment scores (-0.92 to +0.45).
- **slack_alerts.csv:** Datadog APM, PagerDuty, and Cloudflare war room monitoring alerts.
- **Custom CSV Portal:** Upload any enterprise dataset — the ML pipeline auto-adapts in <2 seconds.

6. Local Quickstart & Verification Suite

```
# 1. Clone & install
git clone https://github.com/abhishek9680/CortexKPI.git
cd CortexKPI
pip install -r requirements.txt

# 2. Launch FastAPI Server
python main.py # Dashboard live at http://127.0.0.1:8000

# 3. Run Verification Test Suite
python test_production_api.py
python test_custom_inputs.py
```